Student Performance Analyzer

February 17, 2026

[1]:
```python
import numpy as np
import pandas as pd

# ---------------- Exception Handling ----------------
def safe_input(prompt, cast_type=str):
    while True:
        try:
            value = cast_type(input(prompt))
            return value
        except ValueError:
            print(f"Invalid input! Please enter a valid {cast_type.__name__}.")

# ---------------- OOP Classes ----------------
class Student:
    def __init__(self, name, roll_no, age, class_name, marks):
        self.name = name
        self.roll_no = roll_no
        self.age = age
        self.class_name = class_name
        self.marks = marks  # dictionary of subject: marks

    def display_details(self):
        print(f"\nName: {self.name}, Roll No: {self.roll_no}, "
              f"Age: {self.age}, Class: {self.class_name}")
        print("Marks:", self.marks)

    def calculate_average(self):
        return np.mean(list(self.marks.values()))

    def calculate_grade(self):
        avg = self.calculate_average()
        if avg >= 90:
            return "A"
        elif avg >= 75:
            return "B"
        elif avg >= 60:
            return "C"
```

```
        elif avg >= 40:
            return "D"
        else:
            return "F"


class Classroom:
    def __init__(self):
        self.students = []

    def add_student(self, student):
        self.students.append(student)

    def remove_student(self, roll_no):
        self.students = [s for s in self.students if s.roll_no != roll_no]

    def overall_performance(self):
        if not self.students:
            return None
        all_marks = [s.calculate_average() for s in self.students]
        return {
            "Mean": np.mean(all_marks),
            "Median": np.median(all_marks),
            "Std Dev": np.std(all_marks)
        }

    def to_dataframe(self):
        data = []
        for s in self.students:
            row = {
                "Name": s.name,
                "Roll No": s.roll_no,
                "Age": s.age,
                "Class": s.class_name,
                "Average Marks": s.calculate_average(),
                "Grade": s.calculate_grade()
            }
            row.update(s.marks)  # add subjects
            data.append(row)
        return pd.DataFrame(data)


# ---------------- Main Program ----------------
def main():
    classroom = Classroom()

    print("----- Student Performance Analyzer -----")
```

```
    n = safe_input("Enter number of students: ", int)

    for _ in range(n):
        print("\nEnter details of student:")
        name = safe_input("Name: ", str)
        roll_no = safe_input("Roll No: ", int)
        age = safe_input("Age: ", int)
        class_name = safe_input("Class: ", str)

        marks = {}
        subjects = ["Math", "Science", "English"]
        for sub in subjects:
            marks[sub] = safe_input(f"Enter marks in {sub}: ", int)

        student = Student(name, roll_no, age, class_name, marks)
        classroom.add_student(student)

    # Display all students
    print("\n----- Student Details -----")
    for s in classroom.students:
        s.display_details()
        print("Average:", s.calculate_average(), "Grade:", s.calculate_grade())

    # NumPy Analysis
    print("\n----- NumPy Analysis -----")
    all_marks = np.array([list(s.marks.values()) for s in classroom.students])
    print("Average per subject:", np.mean(all_marks, axis=0))
    print("Highest per subject:", np.max(all_marks, axis=0))
    print("Lowest per subject:", np.min(all_marks, axis=0))
    print("Overall Performance:", classroom.overall_performance())

    # Pandas Analysis
    print("\n----- Pandas Analysis -----")
    df = classroom.to_dataframe()
    print("\nDataFrame:\n", df)
    print("\nTop 5 Students by Avg Marks:\n", df.nlargest(5, "Average Marks"))
    print("\nBottom 5 Students by Avg Marks:\n", df.nsmallest(5, "Average␣
↪Marks"))
    print("\nGroup by Class (Avg Marks):\n",
          df.groupby("Class")["Average Marks"].mean())

    # Save to file
    try:
        df.to_csv("student_report.csv", index=False)
        print("\nReport saved as 'student_report.csv'")
    except Exception as e:
        print("Error saving file:", e)
```

```python
# Run Program
if __name__ == "__main__":
    try:
        main()
    except Exception as e:
        print("Unexpected error:", e)
```

```
----- Student Performance Analyzer -----
Enter number of students: 2

Enter details of student:
Name: Swapnil
Roll No: 100
Age: 22
Class: BSC Computer Science
Enter marks in Math: 91
Enter marks in Science: 90
Enter marks in English: 93

Enter details of student:
Name: Shreyas
Roll No: 101
Age: 22
Class: BSC Computer Science
Enter marks in Math: 89
Enter marks in Science: 91
Enter marks in English: 90

----- Student Details -----

Name: Swapnil , Roll No: 100, Age: 22, Class: BSC Computer Science
Marks: {'Math': 91, 'Science': 90, 'English': 93}
Average: 91.33333333333333 Grade: A

Name: Shreyas, Roll No: 101, Age: 22, Class: BSC Computer Science
Marks: {'Math': 89, 'Science': 91, 'English': 90}
Average: 90.0 Grade: A

----- NumPy Analysis -----
Average per subject: [90.  90.5 91.5]
Highest per subject: [91 91 93]
Lowest per subject: [89 90 90]
Overall Performance: {'Mean': 90.66666666666666, 'Median': 90.66666666666666,
'Std Dev': 0.6666666666666643}

----- Pandas Analysis -----
```

```
DataFrame:
        Name  Roll No  Age                 Class  Average Marks Grade  Math  \
0  Swapnil       100   22  BSC Computer Science      91.333333     A    91
1   Shreyas      101   22  BSC Computer Science      90.000000     A    89

   Science  English
0       90       93
1       91       90

Top 5 Students by Avg Marks:
        Name  Roll No  Age                 Class  Average Marks Grade  Math  \
0  Swapnil       100   22  BSC Computer Science      91.333333     A    91
1   Shreyas      101   22  BSC Computer Science      90.000000     A    89

   Science  English
0       90       93
1       91       90

Bottom 5 Students by Avg Marks:
        Name  Roll No  Age                 Class  Average Marks Grade  Math  \
1   Shreyas      101   22  BSC Computer Science      90.000000     A    89
0  Swapnil       100   22  BSC Computer Science      91.333333     A    91

   Science  English
1       91       90
0       90       93

Group by Class (Avg Marks):
 Class
BSC Computer Science    90.666667
Name: Average Marks, dtype: float64

Report saved as 'student_report.csv'
```

[ ]: